Understanding Machine Learning
Solution Manual

Written by Alon Gonen*
Edited by Dana Rubinstein

November 17, 2014

2 Gentle Start

1. Given $S = ((\mathbf{x}_i, y_i))_{i=1}^m$, define the multivariate polynomial

$$p_S(\mathbf{x}) = -\prod_{i \in [m]: y_i = 1} \|\mathbf{x} - \mathbf{x}_i\|^2 \ .$$

   Then, for every i s.t. $y_i = 1$ we have $p_S(\mathbf{x}_i) = 0$, while for every other $\mathbf{x}$ we have $p_S(\mathbf{x}) < 0$.

2. By the linearity of expectation,

$$\begin{aligned}
\mathop{\mathbb{E}}_{S|_x \sim \mathcal{D}^m}[L_S(h)] &= \mathop{\mathbb{E}}_{S|_x \sim \mathcal{D}^m}\left[\frac{1}{m}\sum_{i=1}^m \mathbb{1}_{[h(x_i) \neq f(x_i)]}\right] \\
&= \frac{1}{m}\sum_{i=1}^m \mathop{\mathbb{E}}_{x_i \sim \mathcal{D}}[\mathbb{1}_{[h(x_i) \neq f(x_i)]}] \\
&= \frac{1}{m}\sum_{i=1}^m \mathop{\mathbb{P}}_{x_i \sim \mathcal{D}}[h(x_i) \neq f(x_i)] \\
&= \frac{1}{m} \cdot m \cdot L_{(\mathcal{D},f)}(h) \\
&= L_{(\mathcal{D},f)}(h) \ .
\end{aligned}$$

*The solutions to Chapters 13,14 were written by Shai Shalev-Shwartz

3. (a) First, observe that by definition, A labels positively all the positive instances in the training set. Second, as we assume realizability, and since the tightest rectangle enclosing all positive examples is returned, all the negative instances are labeled correctly by A as well. We conclude that A is an ERM.

   (b) Fix some distribution $\mathcal{D}$ over $\mathcal{X}$, and define $R^\star$ as in the hint. Let f be the hypothesis associated with $R^\star$ a training set S, denote by $R(S)$ the rectangle returned by the proposed algorithm and by $A(S)$ the corresponding hypothesis. The definition of the algorithm A implies that $R(S) \subseteq R^*$ for every S. Thus,

$$L_{(\mathcal{D},f)}(R(S)) = \mathcal{D}(R^\star \setminus R(S)) \ .$$

   Fix some $\epsilon \in (0,1)$. Define R_1, R_2, R_3 and R_4 as in the hint. For each $i \in [4]$, define the event

$$F_i = \{S|_x : S|_x \cap R_i = \emptyset\} \ .$$

   Applying the union bound, we obtain

$$\mathcal{D}^m(\{S : L_{(\mathcal{D},f)}(A(S)) > \epsilon\}) \leq \mathcal{D}^m\left(\bigcup_{i=1}^{4} F_i\right) \leq \sum_{i=1}^{4} \mathcal{D}^m(F_i) \ .$$

   Thus, it suffices to ensure that $\mathcal{D}^m(F_i) \leq \delta/4$ for every i. Fix some $i \in [4]$. Then, the probability that a sample is in F_i is the probability that all of the instances don't fall in R_i, which is exactly $(1-\epsilon/4)^m$. Therefore,

$$\mathcal{D}^m(F_i) = (1-\epsilon/4)^m \leq \exp(-m\epsilon/4) \ ,$$

   and hence,

$$\mathcal{D}^m(\{S : L_{(\mathcal{D},f)}(A(S)) > \epsilon]\}) \leq 4\exp(-m\epsilon/4) \ .$$

   Plugging in the assumption on m, we conclude our proof.

   (c) The hypothesis class of axis aligned rectangles in $\mathbb{R}^d$ is defined as follows. Given real numbers $a_1 \leq b_1, a_2 \leq b_2, \ldots, a_d \leq b_d$, define the classifier $h_{(a_1,b_1,\ldots,a_d,b_d)}$ by

$$h_{(a_1,b_1,\ldots,a_d,b_d)}(x_1,\ldots,x_d) = \begin{cases} 1 & \text{if } \forall i \in [d],\ a_i \leq x_i \leq b_i \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

The class of all axis-aligned rectangles in $\mathbb{R}^d$ is defined as

$$\mathcal{H}^d_{rec} = \{h_{(a_1,b_1,\dots,a_d,b_d)} : \forall i \in [d],\ a_i \le b_i, \}.$$

It can be seen that the same algorithm proposed above is an ERM for this case as well. The sample complexity is analyzed similarly. The only difference is that instead of 4 strips, we have $2d$ strips (2 strips for each dimension). Thus, it suffices to draw a training set of size $\left\lceil \frac{2d\log(2d/\delta)}{\epsilon} \right\rceil$.

(d) For each dimension, the algorithm has to find the minimal and the maximal values among the positive instances in the training sequence. Therefore, its runtime is $O(md)$. Since we have shown that the required value of m is at most $\left\lceil \frac{2d\log(2d/\delta)}{\epsilon} \right\rceil$, it follows that the runtime of the algorithm is indeed polynomial in $d, 1/\epsilon$, and $\log(1/\delta)$.

3 A Formal Learning Model

1. The proofs follow (almost) immediately from the definition. We will show that the sample complexity is monotonically decreasing in the accuracy parameter ϵ. The proof that the sample complexity is monotonically decreasing in the confidence parameter δ is analogous.

   Denote by $\mathcal{D}$ an unknown distribution over $\mathcal{X}$, and let $f \in \mathcal{H}$ be the target hypothesis. Denote by A an algorithm which learns $\mathcal{H}$ with sample complexity $m_{\mathcal{H}}(\cdot,\cdot)$. Fix some $\delta \in (0,1)$. Suppose that $0 < \epsilon_1 \le \epsilon_2 \le 1$. We need to show that $m_1 \stackrel{\text{def}}{=} m_{\mathcal{H}}(\epsilon_1,\delta) \ge m_{\mathcal{H}}(\epsilon_2,\delta) \stackrel{\text{def}}{=} m_2$. Given an i.i.d. training sequence of size $m \ge m_1$, we have that with probability at least $1-\delta$, A returns a hypothesis h such that

$$L_{\mathcal{D},f}(h) \le \epsilon_1 \le \epsilon_2\ .$$

   By the minimality of m_2, we conclude that $m_2 \le m_1$.

2. (a) We propose the following algorithm. If a positive instance x_+ appears in S, return the (true) hypothesis h_{x_+}. If S doesn't contain any positive instance, the algorithm returns the all-negative hypothesis. It is clear that this algorithm is an ERM.

   (b) Let $\epsilon \in (0,1)$, and fix the distribution $\mathcal{D}$ over $\mathcal{X}$. If the true hypothesis is h^-, then our algorithm returns a perfect hypothesis.

Assume now that there exists a unique positive instance x_+. It's clear that if x_+ appears in the training sequence S, our algorithm returns a perfect hypothesis. Furthermore, if $\mathcal{D}[\{x_+\}] \leq \epsilon$ then in any case, the returned hypothesis has a generalization error of at most ϵ (with probability 1). Thus, it is only left to bound the probability of the case in which $\mathcal{D}[\{x_+\}] > \epsilon$, but x_+ doesn't appear in S. Denote this event by F. Then

$$\mathop{\mathbb{P}}_{S|x \sim \mathcal{D}^m}[F] \leq (1-\epsilon)^m \leq e^{-m\epsilon} .$$

Hence, $\mathcal{H}_{\text{Singleton}}$ is PAC learnable, and its sample complexity is bounded by

$$m_{\mathcal{H}}(\epsilon, \delta) \leq \left\lceil \frac{\log(1/\delta)}{\epsilon} \right\rceil .$$

3. Consider the ERM algorithm A which given a training sequence $S = ((\mathbf{x}_i, y_i))_{i=1}^m$, returns the hypothesis $\hat{h}$ corresponding to the "tightest" circle which contains all the positive instances. Denote the radius of this hypothesis by $\hat{r}$. Assume realizability and let $h^\star$ be a circle with zero generalization error. Denote its radius by $r^\star$.

   Let $\epsilon, \delta \in (0,1)$. Let $\bar{r} \leq r^*$ be a scalar s.t. $\mathcal{D}_{\mathcal{X}}(\{x : \bar{r} \leq \|\mathbf{x}\| \leq r^\star\}) = \epsilon$. Define $E = \{\mathbf{x} \in \mathbb{R}^2 \ : \ \bar{r} \leq \|\mathbf{x}\| \leq r^\star\}$. The probability (over drawing S) that $L_{\mathcal{D}}(h_S) \geq \epsilon$ is bounded above by the probability that no point in S belongs to E. This probability of this event is bounded above by

   $$(1-\epsilon)^m \leq e^{-\epsilon m} .$$

   The desired bound on the sample complexity follows by requiring that $e^{-\epsilon m} \leq \delta$.

4. We first observe that $\mathcal{H}$ is finite. Let us calculate its size accurately. Each hypothesis, besides the all-negative hypothesis, is determined by deciding for each variable x_i, whether x_i, $\bar{x}_i$ or none of which appear in the corresponding conjunction. Thus, $|\mathcal{H}| = 3^d + 1$. We conclude that $\mathcal{H}$ is PAC learnable and its sample complexity can be bounded by

   $$m_{\mathcal{H}}(\epsilon, \delta) \leq \left\lceil \frac{d \log 3 + \log(1/\delta)}{\epsilon} \right\rceil .$$

   Let's describe our learning algorithm. We define $h_0 = x_1 \cap \bar{x}_1 \cap \ldots \cap x_d \cap \bar{x}_d$. Observe that h_0 is the always-minus hypothesis. Let $((\mathbf{a}^1, y^1), \ldots, (\mathbf{a}^m, y^m))$ be an i.i.d. training sequence of size m. Since we cannot produce any information from negative examples, our algorithm neglects them. For each positive example a, we remove from h_i all the literals that are missing in a. That is, if $a_i = 1$, we remove $\bar{x}_i$ from h and if $a_i = 0$, we remove x_i from h_i. Finally, our algorithm returns h_m.

By construction and realizability, h_i labels positively all the positive examples among $\mathbf{a}^1, \ldots, \mathbf{a}^i$. From the same reasons, the set of literals in h_i contains the set of literals in the target hypothesis. Thus, h_i classifies correctly the negative elements among $\mathbf{a}^1, \ldots, \mathbf{a}^i$. This implies that h_m is an ERM.

Since the algorithm takes linear time (in terms of the dimension d) to process each example, the running time is bounded by $O(m \cdot d)$.

5. Fix some $h \in \mathcal{H}$ with $L_{(\overline{\mathcal{D}}_m, f)}(h) > \epsilon$. By definition,

$$\frac{\mathbb{P}_{X\sim\mathcal{D}_1}[h(X) = f(X)] + \ldots + \mathbb{P}_{X\sim\mathcal{D}_m}[h(X) = f(X))]}{m} < 1 - \epsilon \ .$$

We now bound the probability that h is consistent with S (i.e., that $L_S(h) = 0$) as follows:

$$\begin{aligned}
\underset{S\sim\prod_{i=1}^m \mathcal{D}_i}{\mathbb{P}}[L_S(h) = 0] &= \prod_{i=1}^m \underset{X\sim\mathcal{D}_i}{\mathbb{P}}[h(X) = f(X)] \\
&= \left(\left(\prod_{i=1}^m \underset{X\sim\mathcal{D}_i}{\mathbb{P}}[h(X) = f(X)]\right)^{\frac{1}{m}}\right)^m \\
&\leq \left(\frac{\sum_{i=1}^m \mathbb{P}_{X\sim\mathcal{D}_i}[h(X) = f(X)]}{m}\right)^m \\
&< (1-\epsilon)^m \\
&\leq e^{-\epsilon m} \ .
\end{aligned}$$

The first inequality is the geometric-arithmetic mean inequality. Applying the union bound, we conclude that the probability that there exists some $h \in \mathcal{H}$ with $L_{(\overline{\mathcal{D}}_m, f)}(h) > \epsilon$, which is consistent with S is at most $|\mathcal{H}| \exp(-\epsilon m)$.

6. Suppose that $\mathcal{H}$ is agnostic PAC learnable, and let A be a learning algorithm that learns $\mathcal{H}$ with sample complexity $m_{\mathcal{H}}(\cdot, \cdot)$. We show that $\mathcal{H}$ is PAC learnable using A.

Let $\mathcal{D}, f$ be an (unknown) distribution over $\mathcal{X}$, and the target function respectively. We may assume w.l.o.g. that $\mathcal{D}$ is a joint distribution over $\mathcal{X} \times \{0,1\}$, where the conditional probability of y given x is determined deterministically by f. Since we assume realizability, we have $\inf_{h \in \mathcal{H}} L_{\mathcal{D}}(h) = 0$. Let $\epsilon, \delta \in (0,1)$. Then, for every positive integer $m \geq m_{\mathcal{H}}(\epsilon, \delta)$, if we equip A with a training set S consisting of m i.i.d. instances which are labeled by f, then with probability at least $1-\delta$ (over the choice of $S|_x$), it returns a hypothesis h with

$$\begin{aligned} L_{\mathcal{D}}(h) &\leq \inf_{h' \in \mathcal{H}} L_{\mathcal{D}}(h') + \epsilon \\ &= 0 + \epsilon \\ &= \epsilon \ . \end{aligned}$$

7. Let $x \in \mathcal{X}$. Let α_x be the conditional probability of a positive label given x. We have

$$\begin{aligned} \mathbb{P}[f_{\mathcal{D}}(X) \neq y | X = x] &= \mathbb{1}_{[\alpha_x \geq 1/2]} \cdot \mathbb{P}[Y = 0 | X = x] + \mathbb{1}_{[\alpha_x < 1/2]} \cdot \mathbb{P}[Y = 1 | X = x] \\ &= \mathbb{1}_{[\alpha_x \geq 1/2]} \cdot (1 - \alpha_x) + \mathbb{1}_{[\alpha_x < 1/2]} \cdot \alpha_x \\ &= \min\{\alpha_x, 1 - \alpha_x\}. \end{aligned}$$

Let g be a classifier[1] from $\mathcal{X}$ to $\{0,1\}$. We have

$$\begin{aligned} \mathbb{P}[g(X) \neq Y | X = x] &= \mathbb{P}[g(X) = 0 | X = x] \cdot \mathbb{P}[Y = 1 | X = x] \\ &+ \mathbb{P}[g(X) = 1 | X = x] \cdot \mathbb{P}[Y = 0 | X = x] \\ &= \mathbb{P}[g(X) = 0 | X = x] \cdot \alpha_x + \mathbb{P}[g(X) = 1 | X = x] \cdot (1 - \alpha_x) \\ &\geq \mathbb{P}[g(X) = 0 | X = x] \cdot \min\{\alpha_x, 1 - \alpha_x\} \\ &+ \mathbb{P}[g(X) = 1 | x] \cdot \min\{\alpha_x, 1 - \alpha_x\} \\ &= \min\{\alpha_x, 1 - \alpha_x\}, \end{aligned}$$

The statement follows now due to the fact that the above is true for every $x \in \mathcal{X}$. More formally, by the law of total expectation,

$$\begin{aligned} L_{\mathcal{D}}(f_{\mathcal{D}}) &= \mathbb{E}_{(x,y) \sim \mathcal{D}}[\mathbb{1}_{[f_{\mathcal{D}}(x) \neq y]}] \\ &= \mathbb{E}_{x \sim \mathcal{D}_X}\left[\mathbb{E}_{y \sim \mathcal{D}_{Y|x}}[\mathbb{1}_{[f_{\mathcal{D}}(x) \neq y]} | X = x]\right] \\ &= \mathbb{E}_{x \sim \mathcal{D}_X}[\alpha_x] \\ &\leq \mathbb{E}_{x \sim \mathcal{D}_X}\left[\mathbb{E}_{y \sim \mathcal{D}_{Y|x}}[\mathbb{1}_{[g(x) \neq y]} | X = x]\right] \\ &= L_{\mathcal{D}}(g) \ . \end{aligned}$$

[1] As we shall see, g might be non-deterministic.

8. (a) This was proved in the previous exercise.

   (b) We proved in the previous exercise that for every distribution $\mathcal{D}$, the bayes optimal predictor $f_{\mathcal{D}}$ is optimal w.r.t. $\mathcal{D}$.

   (c) Choose any distribution $\mathcal{D}$. Then A is not better than $f_{\mathcal{D}}$ w.r.t. $\mathcal{D}$.

9. (a) Suppose that $\mathcal{H}$ is PAC learnable in the one-oracle model. Let A be an algorithm which learns $\mathcal{H}$ and denote by $m_{\mathcal{H}}$ the function that determines its sample complexity. We prove that $\mathcal{H}$ is PAC learnable also in the two-oracle model.

   Let $\mathcal{D}$ be a distribution over $\mathcal{X} \times \{0,1\}$. Note that drawing points from the negative and positive oracles with equal provability is equivalent to obtaining i.i.d. examples from a distribution $\mathcal{D}'$ which gives equal probability to positive and negative examples. Formally, for every subset $E \subseteq \mathcal{X}$ we have

$$\mathcal{D}'[E] = \frac{1}{2}\mathcal{D}^+[E] + \frac{1}{2}\mathcal{D}^-[E].$$

   Thus, $\mathcal{D}'[\{x : f(x) = 1\}] = \mathcal{D}'[\{x : f(x) = 0\}] = \frac{1}{2}$. If we let A an access to a training set which is drawn i.i.d. according to $\mathcal{D}'$ with size $m_{\mathcal{H}}(\epsilon/2, \delta)$, then with probability at least $1 - \delta$, A returns h with

$$\begin{aligned}
\epsilon/2 \geq L_{(\mathcal{D}',f)}(h) &= \underset{x\sim\mathcal{D}'}{\mathbb{P}}[h(x) \neq f(x)] \\
&= \underset{x\sim\mathcal{D}'}{\mathbb{P}}[f(x) = 1, h(x) = 0] + \underset{x\sim\mathcal{D}'}{\mathbb{P}}[f(x) = 0, h(x) = 1] \\
&= \underset{x\sim\mathcal{D}'}{\mathbb{P}}[f(x) = 1] \cdot \underset{x\sim\mathcal{D}'}{\mathbb{P}}[h(x) = 0 | f(x) = 1] \\
&+ \underset{x\sim\mathcal{D}'}{\mathbb{P}}[f(x) = 0] \cdot \underset{x\sim\mathcal{D}'}{\mathbb{P}}[h(x) = 1 | f(x) = 0] \\
&= \underset{x\sim\mathcal{D}'}{\mathbb{P}}[f(x) = 1] \cdot \underset{x\sim\mathcal{D}}{\mathbb{P}}[h(x) = 0 | f(x) = 1] \\
&+ \underset{x\sim\mathcal{D}'}{\mathbb{P}}[f(x) = 0] \cdot \underset{x\sim\mathcal{D}}{\mathbb{P}}[h(x) = 1 | f(x) = 0] \\
&= \frac{1}{2} \cdot L_{(\mathcal{D}^+,f)}(h) + \frac{1}{2} \cdot L_{(\mathcal{D}^-,f)}(h).
\end{aligned}$$

   This implies that with probability at least $1 - \delta$, both

$$L_{(\mathcal{D}^+,f)}(h) \leq \epsilon \text{ and } L_{(\mathcal{D}^-,f)}(h) \leq \epsilon.$$

Our definition for PAC learnability in the two-oracle model is satisfied. We can bound both $m^+_{\mathcal{H}}(\epsilon, \delta)$ and $m^-_{\mathcal{H}}(\epsilon, \delta)$ by $m_{\mathcal{H}}(\epsilon/2, \delta)$.

(b) Suppose that $\mathcal{H}$ is PAC learnable in the two-oracle model and let A be an algorithm which learns $\mathcal{H}$. We show that $\mathcal{H}$ is PAC learnable also in the standard model.

Let $\mathcal{D}$ be a distribution over $\mathcal{X}$, and denote the target hypothesis by f. Let $\alpha = \mathcal{D}[\{x : f(x) = 1\}]$. Let $\epsilon, \delta \in (0, 1)$. According to our assumptions, there exist $m^+ \stackrel{\text{def}}{=} m^+_{\mathcal{H}}(\epsilon, \delta/2), m^- \stackrel{\text{def}}{=} m^-_{\mathcal{H}}(\epsilon, \delta/2)$ s.t. if we equip A with m^+ examples drawn i.i.d. from $\mathcal{D}^+$ and m^- examples drawn i.i.d. from $\mathcal{D}^-$, then, with probability at least $1 - \delta/2$, A will return h with

$$L_{(\mathcal{D}^+,f)}(h) \leq \epsilon \wedge L_{(\mathcal{D}^-,f)}(h) \leq \epsilon \ .$$

Our algorithm B draws $m = \max\{2m^+/\epsilon, 2m^-/\epsilon, \frac{8\log(4/\delta)}{\epsilon}\}$ samples according to $\mathcal{D}$. If there are less then m^+ positive examples, B returns h^-. Otherwise, if there are less then m^- negative examples, B returns h^+. Otherwise, B runs A on the sample and returns the hypothesis returned by A.

First we observe that if the sample contains m^+ positive instances and m^- negative instances, then the reduction to the two-oracle model works well. More precisely, with probability at least $1 - \delta/2$, A returns h with

$$L_{(\mathcal{D}^+,f)}(h) \leq \epsilon \wedge L_{(\mathcal{D}^-,f)}(h) \leq \epsilon \ .$$

Hence, with probability at least $1 - \delta/2$, the algorithm B returns (the same) h with

$$L_{(\mathcal{D},f)}(h) = \alpha \cdot L_{(\mathcal{D}^+,f)}(h) + (1 - \alpha) \cdot L_{(\mathcal{D}^-,f)}(h) \leq \epsilon$$

We consider now the following cases:

- Assume that both $\alpha \geq \epsilon$. We show that with probability at least $1 - \delta/4$, the sample contain m^+ positive instances. For each $i =\in [m]$, define the indicator random variable Z_i, which gets the value 1 iff the i-th element in the sample is positive. Define $Z = \sum_{i=1}^m Z_i$ to be the number of positive examples that were drawn. Clearly, $\mathbb{E}[Z] = \alpha m$. Using Chernoff bound, we obtain

$$\mathbb{P}[Z < (1 - \frac{1}{2})\alpha m] < e^{\frac{-m\alpha}{8}} \ .$$

By the way we chose m, we conclude that

$$\mathbb{P}[Z < m_+] < \delta/4 \ .$$

Similarly, if $1-\alpha \geq \epsilon$, the probability that less than m^- negative examples were drawn is at most $\delta/4$. If both $\alpha \geq \epsilon$ and $1-\alpha \geq \epsilon$, then, by the union bound, with probability at least $1-\delta/2$, the training set contains at least m^+ and m^- positive and negative instances respectively. As we mentioned above, if this is the case, the reduction to the two-oracle model works with probability at least $1-\delta/2$. The desired conclusion follows by applying the union bound.

- Assume that $\alpha < \epsilon$, and less than m^+ positive examples are drawn. In this case, B will return the hypothesis h^-. We obtain

$$L_{\mathcal{D}}(h) = \alpha < \epsilon \ .$$

Similarly, if $(1-\alpha) < \epsilon$, and less than m^- negative examples are drawn, B will return h^+. In this case,

$$L_{\mathcal{D}}(h) = 1-\alpha < \epsilon \ .$$

All in all, we have shown that with probability at least $1-\delta$, B returns a hypothesis h with $L_{(\mathcal{D},f)}(h) < \epsilon$. This satisfies our definition for PAC learnability in the one-oracle model.

4 Learning via Uniform Convergence

1. (a) Assume that for every $\epsilon, \delta \in (0,1)$, and every distribution $\mathcal{D}$ over $\mathcal{X} \times \{0,1\}$, there exists $m(\epsilon,\delta) \in \mathbb{N}$ such that for every $m \geq m(\epsilon,\delta)$,

$$\underset{S\sim\mathcal{D}^m}{\mathbb{P}}[L_{\mathcal{D}}(A(S)) > \epsilon] < \delta \ .$$

Let $\lambda > 0$. We need to show that there exists $m_0 \in \mathbb{N}$ such that for every $m \geq m_0$, $\mathbb{E}_{S\sim\mathcal{D}^m}[L_{\mathcal{D}}(A(S))] \leq \lambda$. Let $\epsilon = \min(1/2, \lambda/2)$. Set $m_0 = m_{\mathcal{H}}(\epsilon,\epsilon)$. For every $m \geq m_0$, since the loss is bounded above by 1, we have

$$\begin{aligned}
\mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}[L_{\mathcal{D}}(A(S))] &\le \mathop{\mathbb{P}}_{S\sim\mathcal{D}^m}[L_{\mathcal{D}}(A(S)) > \lambda/2]\cdot 1 + \mathop{\mathbb{P}}_{S\sim\mathcal{D}^m}[L_{\mathcal{D}}(A(S)) \le \lambda/2]\cdot \lambda/2 \\
&\le \mathop{\mathbb{P}}_{S\sim\mathcal{D}^m}[L_{\mathcal{D}}(A(S)) > \epsilon] + \lambda/2 \\
&\le \epsilon + \lambda/2 \\
&\le \lambda/2 + \lambda/2 \\
&= \lambda\ .
\end{aligned}$$

(b) Assume now that

$$\lim_{m\to\infty} \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}[L_{\mathcal{D}}(A(S))] = 0\ .$$

Let $\epsilon, \delta \in (0,1)$. There exists some $m_0 \in \mathbb{N}$ such that for every $m \ge m_0$, $\mathbb{E}_{S\sim\mathcal{D}^m}[L_{\mathcal{D}}(A(S))] \le \epsilon\cdot\delta$. By Markov's inequality,

$$\begin{aligned}
\mathop{\mathbb{P}}_{S\sim\mathcal{D}^m}[L_{\mathcal{D}}(A(S)) > \epsilon] &\le \frac{\mathbb{E}_{S\sim\mathcal{D}^m}[L_{\mathcal{D}}(A(S))]}{\epsilon} \\
&\le \frac{\epsilon\delta}{\epsilon} \\
&= \delta\ .
\end{aligned}$$

2. The left inequality follows from Corollary 4.4. We prove the right inequality. Fix some $h \in \mathcal{H}$. Applying Hoeffding's inequality, we obtain

$$\mathop{\mathbb{P}}_{S\sim\mathcal{D}^m}[|L_{\mathcal{D}}(h) - L_S(h)| \ge \epsilon/2] \le 2\exp\left(-\frac{2m\epsilon^2}{(b-a)^2}\right)\ . \tag{2}$$

The desired inequality is obtained by requiring that the right-hand side of Equation (2) is at most $\delta/|\mathcal{H}|$, and then applying the union bound.

5 The Bias-Complexity Tradeoff

1. We simply follow the hint. By Lemma B.1,

$$\begin{aligned}
\mathop{\mathbb{P}}_{S\sim\mathcal{D}^m}[L_{\mathcal{D}}(A(S)) \ge 1/8] &= \mathop{\mathbb{P}}_{S\sim\mathcal{D}^m}[L_{\mathcal{D}}(A(S)) \ge 1 - 7/8] \\
&\ge \frac{\mathbb{E}[L_{\mathcal{D}}(A(S))] - (1 - 7/8)}{7/8} \\
&\ge \frac{1/8}{7/8} \\
&= 1/7\ .
\end{aligned}$$

2. We will provide a qualitative answer. In the next section, we'll be able to quantify our argument.

   Denote by $\mathcal{H}_d$ the class of axis-aligned rectangles in $\mathbb{R}^d$. Since $\mathcal{H}_5 \supseteq \mathcal{H}_2$, the approximation error of the class $\mathcal{H}_5$ is smaller. However, the complexity of $\mathcal{H}_5$ is larger, so we expect its estimation error to be larger. So, if we have a limited amount of training examples, we would prefer to learn the smaller class.

3. We modify the proof of the NFL theorem, based on the assumption that $|\mathcal{X}| \geq km$ (while referring to some equations from the book). Choose C to be a set with size km. Then, Equation (5.5) (in the book) becomes

$$L_{\mathcal{D}_i}(h) = \frac{1}{km} \sum_{x \in C} \mathbb{1}_{[h(x) \neq f_i(x)]} \geq \frac{1}{km} \sum_{r=1}^{p} \mathbb{1}_{[h(v_r) \neq f_i(v_r)]} \geq \frac{k-1}{pk} \sum_{r=1}^{p} \mathbb{1}_{[h(v_r) \neq f_i(v_r)]} \,.$$

   Consequently, the right-hand side of Equation (5.6) becomes

$$\frac{k-1}{k} \min_{r \in [p]} \frac{1}{T} \sum_{i=1}^{T} \mathbb{1}_{[A(S_j^i)(v_r) \neq f_i(v_r)]}$$

   It follows that

$$\mathbb{E}[L_{\mathcal{D}}(A(S))] \geq \frac{k-1}{2k} \,.$$

6 The VC-dimension

1. Let $\mathcal{H}' \subseteq \mathcal{H}$ be two hypothesis classes for binary classification. Since $\mathcal{H}' \subseteq \mathcal{H}$, then for every $C = \{c_1, \ldots, c_m\} \subseteq \mathcal{X}$, we have $\mathcal{H}'_C \subseteq \mathcal{H}_C$. In particular, if C is shattered by $\mathcal{H}'$, then C is shattered by $\mathcal{H}$ as well. Thus, $\mathrm{VCdim}(\mathcal{H}') \leq \mathrm{VCdim}(\mathcal{H})$.

2. (a) We claim that $\mathrm{VCdim}(\mathcal{H}_{=k}) = \min\{k, |\mathcal{X}| - k\}$. First, we show that $\mathrm{VCdim}(\mathcal{H}_{=k}) \leq k$. Let $C \subseteq \mathcal{X}$ be a set of size $k+1$. Then, there doesn't exist $h \in \mathcal{H}_{=k}$ which satisfies $h(x) = 1$ for all $x \in C$. Analogously, if $C \subseteq \mathcal{X}$ is a set of size $|\mathcal{X}| - k + 1$, there is no $h \in \mathcal{H}_{=k}$ which satisfies $h(x) = 0$ for all $x \in C$. Hence, $\mathrm{VCdim}(\mathcal{H}_{=k}) \leq \min\{k, |\mathcal{X}| - k\}$.

   It's left to show that $\mathrm{VCdim}(\mathcal{H}_{=k}) \geq \min\{k, |\mathcal{X}| - k\}$. Let $C = \{x_1, \ldots, x_m\} \subseteq \mathcal{X}$ be a set with of size $m \leq \min\{k, |\mathcal{X}| - k\}$. Let $(y_1, \ldots, y_m) \in \{0,1\}^m$ be a vector of labels. Denote $\sum_{i=1}^m y_i$

by s. Pick an arbitrary subset $E \subseteq \mathcal{X} \setminus C$ of $k - s$ elements, and let $h \in \mathcal{H}_{=k}$ be the hypothesis which satisfies $h(x_i) = y_i$ for every $x_i \in C$, and $h(x) = \mathbb{1}_{[E]}$ for every $x \in \mathcal{X} \setminus C$. We conclude that C is shattered by $\mathcal{H}_{=k}$. It follows that $\text{VCdim}(\mathcal{H}_{=k}) \geq \min\{k, |\mathcal{X}| - k\}$.

(b) We claim that $\text{VCdim}(\mathcal{H}_{\leq k}) = k$. First, we show that $\text{VCdim}(\mathcal{H}^{\mathcal{X}}_{\leq k}) \leq k$. Let $C \subseteq \mathcal{X}$ be a set of size $k + 1$. Then, there doesn't exist $h \in \mathcal{H}_{\leq k}$ which satisfies $h(x) = 1$ for all $x \in C$.
It's left to show that $\text{VCdim}(\mathcal{H}_{\leq k}) \geq k$. Let $C = \{x_1, \ldots, x_m\} \subseteq \mathcal{X}$ be a set with of size $m \leq k$. Let $(y_1, \ldots, y_m) \in \{0, 1\}^m$ be a vector of labels. This labeling is obtained by some hypothesis $h \in \mathcal{H}_{\leq k}$ which satisfies $h(x_i) = y_i$ for every $x_i \in C$, and $h(x) = 0$ for every $x \in \mathcal{X} \setminus C$. We conclude that C is shattered by $\mathcal{H}_{\leq k}$. It follows that $\text{VCdim}(\mathcal{H}_{\leq k}) \geq k$.

3. We claim that VC-dimension of $\mathcal{H}_{n\text{-parity}}$ is n. First, we note that $|\mathcal{H}_{n\text{-parity}}| = 2^n$. Thus,

$$\text{VCdim}(\mathcal{H}_{n\text{-parity}}) \leq \log(|\mathcal{H}_{n\text{-parity}}|) = n \ .$$

We will conclude the tightness of this bound by showing that the standard basis $\{\mathbf{e}_j\}_{j=1}^n$ is shattered by $\mathcal{H}_{n\text{-parity}}$. Given a vector of labels $(y_1, \ldots, y_n) \in \{0, 1\}^n$, let $J = \{j \in [n] \ : \ y_j = 1\}$. Then $h_J(e_j) = y_j$ for every $j \in [n]$.

4. Let $\mathcal{X} = \mathbb{R}^d$. We will demonstrate all the 4 combinations using hypothesis classes defined over $\mathcal{X} \times \{0, 1\}$. Remember that the empty set is always considered to be shattered.

- $(<, =)$: Let $d \geq 2$ and consider the class $\mathcal{H} = \{\mathbb{1}_{[\|x\|_2 \leq r]} : r \geq 0\}$ of concentric balls. The VC-dimension of this class is 1. To see this, we first observe that if $\mathbf{x} \neq (0, \ldots, 0)$, then $\{\mathbf{x}\}$ is shattered. Second, if $\|\mathbf{x}_1\|_2 \leq \|\mathbf{x}_2\|_2$, then the labeling $y_1 = 0, y_2 = 1$ is not obtained by any hypothesis in $\mathcal{H}$. Let $A = \{\mathbf{e}_1, \mathbf{e}_2\}$, where $\mathbf{e}_1, \mathbf{e}_2$ are the first two elements of the standard basis of $\mathbb{R}^d$. Then, $\mathcal{H}_A = \{(0, 0), (1, 1)\}$, $\{B \subseteq A : \mathcal{H} \text{ shatters } B\} = \{\emptyset, \{\mathbf{e}_1\}, \{\mathbf{e}_2\}\}$, and $\sum_{i=0}^{d} \binom{|A|}{i} = 3$.
- $(=, <)$: Let $\mathcal{H}$ be the class of axis-aligned rectangles in $\mathbb{R}^2$. We have seen that the VC-dimension of $\mathcal{H}$ is 4. Let $A = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$, where $\mathbf{x}_1 = (0, 0), \mathbf{x}_2 = (1, 0), \mathbf{x}_3 = (2, 0)$. All the labelings except $(1, 0, 1)$ are obtained. Thus, $|\mathcal{H}_A| = 7$, $|\{B \subseteq A : \mathcal{H} \text{ shatters } B\}| = 7$, and $\sum_{i=0}^{d} \binom{|A|}{i} = 8$.

- $(<,<)$: Let $d \geq 3$ and consider the class $\mathcal{H} = \{\text{sign}\langle w, x\rangle : w \in \mathbb{R}^d\}$[2] of homogenous halfspaces (see Chapter 9). We will prove in Theorem 9.2 that the VC-dimension of this class is d. However, here we will only rely on the fact that $\text{VCdim}(\mathcal{H}) \geq 3$. This fact follows by observing that the set $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is shattered. Let $A = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$, where $\mathbf{x}_1 = \mathbf{e}_1, \mathbf{x}_2 = \mathbf{e}_2$, and $\mathbf{x}_3 = (1, 1, 0, \ldots, 0)$. Note that all the labelings except $(1, 1, -1)$ and $(-1, -1, 1)$ are obtained. It follows that $|\mathcal{H}_A| = 6$, $|\{B \subseteq A : \mathcal{H} \text{ shatters } B\}| = 7$, and $\sum_{i=0}^{d} \binom{|A|}{i} = 8$.
- $(=,=)$: Let $d = 1$, and consider the class $\mathcal{H} = \{\mathbb{1}_{[x \geq t]} : t \in \mathbb{R}\}$ of thresholds on the line. We have seen that every singleton is shattered by $\mathcal{H}$, and that every set of size at least 2 is not shattered by $\mathcal{H}$. Choose any finite set $A \subseteq \mathbb{R}$. Then each of the three terms in "Sauer's inequality" equals $|A| + 1$.

5. Our proof is a straightforward generalization of the proof in the 2-dimensional case.

   Let us first define the class formally. Given real numbers $a_1 \leq b_1, a_2 \leq b_2, \ldots, a_d \leq b_d$, define the classifier $h_{(a_1,b_1,\ldots,a_d,b_d)}$ by $h_{(a_1,b_1,\ldots,a_d,b_d)}(x_1, \ldots, x_d) = \prod_{i=1}^{d} \mathbb{1}_{[x_i \in [a_i,b_i]]}$. The class of all axis-aligned rectangles in $\mathbb{R}^d$ is defined as $\mathcal{H}^d_{rec} = \{h_{(a_1,b_1,\ldots,a_d,b_d)} : \forall i \in [d],\ a_i \leq b_i, \}$.

   Consider the set $\{\mathbf{x}_1, \ldots, \mathbf{x}_{2d}\}$, where $\mathbf{x}_i = \mathbf{e}_i$ if $i \in [d]$, and $\mathbf{x}_i = -e_{i-d}$ if $i > d$. As in the 2-dimensional case, it's not hard to see that it's shattered. Indeed, let $(y_1, \ldots, y_{2d}) \in \{0, 1\}^{2d}$. Choose $a_i = -2$ if $y_{i+d} = 1$, and $a_i = 0$ otherwise. Similarly, choose $b_i = 2$ if $y_i = 1$, and $b_i = 0$ otherwise. Then $h_{a_1,b_1,\ldots,a_d,b_d}(\mathbf{x}_i) = y_i$ for every $i \in [2d]$. We just proved that $\text{VCdim}(\mathcal{H}^d_{rec}) \geq 2d$.

   Let C be a set of size at least $2d + 1$. We finish our proof by showing that C is not shattered. By the pigeonhole principle, there exists an element $\mathbf{x} \in C$, s.t. for every $j \in [d]$, there exists $\mathbf{x}' \in C$ with $x'_j \leq x_j$, and similarly there exists $\mathbf{x}'' \in C$ with $x''_j \geq x_j$. Thus the labeling in which x is negative, and the rest of the elements in C are positive can not be obtained.

6. (a) Each hypothesis, besides the all-negative hypothesis, is determined by deciding for each variable x_i, whether x_i, $\bar{x}_i$ or none of which appear in the corresponding conjunction. Thus, $|\mathcal{H}^d_{con}| = 3^d + 1$.

[2] We adopt the convention $\text{sign}(0) = 1$.

(b)
$$\mathrm{VCdim}(\mathcal{H}^d_{con}) \le \lfloor \log(|\mathcal{H}^d_{con}|) \rfloor \le 3 \log d \ .$$

(c) We prove that $\mathcal{H}^d_{con} \ge d$ by showing that the set $C = \{\mathbf{e}_j\}_{j=1}^d$ is shattered by $\mathcal{H}^d_{con}$. Let $J \subseteq [d]$ be a subset of indices. We will show that the labeling in which exactly the elements $\{\mathbf{e}_j\}_{j \in J}$ are positive is obtained. If $J = [d]$, pick the all-positive hypothesis h_{empty}. If $J = \emptyset$, pick the all-negative hypothesis $x_1 \wedge \bar{x}_1$. Assume now that $\emptyset \subsetneq J \subsetneq [d]$. Let h be the hypothesis which corresponds to the boolean conjunction $\bigwedge_{j \in J} x_j$. Then, $h(\mathbf{e}_j) = 1$ if $j \in J$, and $h(\mathbf{e}_j) = 0$ otherwise.

(d) Assume by contradiction that there exists a set $C = \{c_1, \ldots, c_{d+1}\}$ for which $|\mathcal{H}_C| = 2^{d+1}$. Define $h_1, \ldots, h_{d+1}$, and $\ell_1, \ldots, \ell_{d+1}$ as in the hint. By the Pigeonhole principle, among $\ell_1, \ldots, \ell_{d+1}$, at least one variable occurs twice. Assume w.l.o.g. that ℓ_1 and ℓ_2 correspond to the same variable. Assume first that $\ell_1 = \ell_2$. Then, ℓ_1 is true on c_1 since ℓ_2 is true on c_1. However, this contradicts our assumptions. Assume now that $\ell_1 \neq \ell_2$. In this case $h_1(c_3)$ is negative, since ℓ_2 is positive on c_3. This again contradicts our assumptions.

(e) First, we observe that $|\mathcal{H}'| = 2^d + 1$. Thus,
$$\mathrm{VCdim}(\mathcal{H}') \le \lfloor \log(|\mathcal{H}|) \rfloor = d \ .$$
We will complete the proof by exhibiting a shattered set with size d. Let
$$C = \{(1,1,\ldots,1) - e_j\}_{j=1}^d = \{(0,1,\ldots,1), \ldots, (1,\ldots,1,0)\} \ .$$
Let $J \subseteq [d]$ be a subset of indices. We will show that the labeling in which exactly the elements $\{(1,1,\ldots,1) - e_j\}_{j \in J}$ are negative is obtained. Assume for the moment that $J \neq \emptyset$. Then the labeling is obtained by the boolean conjunction $\bigwedge_{j \in J} x_j$. Finally, if $J = \emptyset$, pick the all-positive hypothesis $h_\emptyset$.

7. (a) The class $\mathcal{H} = \{\mathbb{1}_{[x \ge t]} : t \in \mathbb{R}\}$ (defined over any non-empty subset of $\mathbb{R}$) of thresholds on the line is infinite, while its VC-dimension equals 1.

(b) The hypothesis class $\mathcal{H} = \{\mathbb{1}_{[x \le 1]}, \mathbb{1}_{[x \le 1/2]}\}$ satisfies the requirements.

8. We will begin by proving the lemma provided in the question:

$$
\begin{aligned}
sin(2^m \pi x) \;&=\; \sin(2^m\pi(0.x_1x_2\ldots)) \\
&= \sin(2\pi(x_1x_2\ldots x_{m-1}.x_mx_{m+1}\ldots)) \\
&= \sin\left(2\pi(x_1x_2\ldots x_{m-1}.x_mx_{m+1}\ldots) - 2\pi(x_1x_2\ldots x_{m-1}.0)\right) \\
&= \sin(2\pi(0.x_mx_{m+1}\ldots).
\end{aligned}
$$

Now, if $x_m = 0$, then $2\pi(0.x_mx_{m+1}\ldots) \in (0,\pi)$ (we use here the fact that $\exists k \geq m$ s.t. $x_k = 1$) , so the expression above is positive, and $\lceil sin(2^m\pi x)\rceil = 1$. On the other hand, if $x_m = 1$, then $2\pi(0.x_mx_{m+1}) \in [\pi, 2\pi)$, so the expression above is non-positive, and $\lceil sin(2^m\pi x)\rceil = 0$. In conclusion, we have that $\lceil sin(2^m\pi x)\rceil = 1 - x_m$.

To prove $VC(\mathcal{H}) = \infty$, we need to pick n points which are shattered by $\mathcal{H}$, for any n. To do so, we construct n points $x_1,\ldots,x_n \in [0,1]$, such that the set of the m-th bits in the binary expansion, as m ranges from 1 to 2^n, ranges over all possible labelings of $x_1,\ldots,x_n$:

$$
\begin{array}{rcccccccccc}
x_1 & = & 0\,. & 0 & 0 & 0 & 0 & \cdots & 1 & 1 \\
x_2 & = & 0\,. & 0 & 0 & 0 & 0 & \cdots & 1 & 1 \\
 & & & & & \vdots & & & & \\
x_{n-1} & = & 0\,. & 0 & 0 & 1 & 1 & \cdots & 1 & 1 \\
x_n & = & 0\,. & 0 & 1 & 0 & 1 & \cdots & 0 & 1
\end{array}
$$

For example, to give the labeling 1 for all instances, we just pick $h(x) = \lceil sin(2^1x)\rceil$, which returns the first bit (column) in the binary expansion. If we wish to give the labeling 1 for $x_1,\ldots,x_{n-1}$, and the labeling 0 for x_n, we pick $h(x) = \lceil sin(2^2x)\rceil$, which returns the 2nd bit in the binary expansion, and so on. We conclude that $x_1,\ldots,x_n$ can be given any labeling by some $h \in \mathcal{H}$, so it is shattered. This can be done for any n, so $\text{VCdim}(\mathcal{H}) = \infty$.

9. We prove that $\text{VCdim}(\mathcal{H}) = 3$. Choose $C = \{1,2,3\}$. The following table shows that C is shattered by $\mathcal{H}$.

1	2	3	a	b	s
-	-	-	0.5	3.5	-1
-	-	+	2.5	3.5	1
-	+	-	1.5	2.5	1
-	+	+	1.5	3.5	1
+	-	-	0.5	1.5	1
+	-	+	1.5	2.5	-1
+	+	-	0.5	2.5	1
+	+	+	0.5	3.5	1

We conclude that $\text{VCdim}(\mathcal{H}) \geq 3$. Let $C = \{x_1, x_2, x_3, x_4\}$ and assume w.l.o.g. that $x_1 < x_2 < x_3 < x_4$ Then, the labeling $y_1 = y_3 = -1$, $y_2 = y_4 = 1$ is not obtained by any hypothesis in $\mathcal{H}$. Thus, $\text{VCdim}(\mathcal{H}) \leq 3$.

10. (a) We may assume that $m < d$, since otherwise the statement is meaningless. Let C be a shattered set of size d. We may assume w.l.o.g. that $\mathcal{X} = C$ (since we can always choose distributions which are concentrated on C). Note that $\mathcal{H}$ contains all the functions from C to $\{0, 1\}$. According to Exercise 3 in Section 5, for every algorithm, there exists a distribution $\mathcal{D}$, for which $\min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) = 0$, but $\mathbb{E}[L_{\mathcal{D}}(A(S))] \geq \frac{k-1}{2k} \underbrace{=}_{k=\frac{d}{m}} \frac{d-m}{2d}$.

    (b) Assume that $\text{VCdim}(\mathcal{H}) = \infty$. Let $\mathcal{A}$ be a learning algorithm. We show that A fails to (PAC) learn $\mathcal{H}$. Choose $\epsilon = \frac{1}{16}$, $\delta = \frac{1}{14}$. For any $m \in \mathbb{N}$, there exists a shattered set of size $d = 2m$. Applying the above, we obtain that there exists a distribution $\mathcal{D}$ for which $\min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) = 0$, but $\mathbb{E}[L_{\mathcal{D}}(A(S))] \geq 1/4$. Applying Lemma B.1 yields that with probability at least $1/7 > \delta$, $L_{\mathcal{D}}(A(S)) - \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) = L_{\mathcal{D}}(A(S) \geq 1/8 > \epsilon$.

11. (a) We may assume w.l.o.g. that for each $i \in [r]$, $\text{VCdim}(\mathcal{H}_i) = d \geq 3$. Let $\mathcal{H} = \bigcup_{i=1}^{r} \mathcal{H}_i$. Let $k \in [d]$, such that $\tau_{\mathcal{H}}(k) = 2^k$. We will show that $k \leq 4d \log(2d) + 2 \log r$.

    By definition of the growth function, we have

$$\tau_{\mathcal{H}}(k) \leq \sum_{i=1}^{r} \tau_{\mathcal{H}_i}(k) \ .$$

    Since $d \geq 3$, by applying Sauer's lemma on each of the terms $\tau_{\mathcal{H}_i}$, we obtain

$$\tau_{\mathcal{H}}(k) < rm^d \ .$$

It follows that $k < d \log m + \log r$. Lemma A.2 implies that $k < 4d \log(2d) + 2 \log r$.

(b) A direct application of the result above yields a weaker bound. We need to employ a more careful analysis.

As before, we may assume w.l.o.g. that $\mathrm{VCdim}(\mathcal{H}_1) = \mathrm{VCdim}(\mathcal{H}_2) = d$. Let $\mathcal{H} = \mathcal{H}_1 \cup \mathcal{H}_2$. Let k be a positive integer such that $k \geq 2d + 2$. We show that $\tau_{\mathcal{H}}(k) < 2^k$. By Sauer's lemma,

$$\begin{aligned}
\tau_{\mathcal{H}}(k) &\leq \tau_{\mathcal{H}_1}(k) + \tau_{\mathcal{H}_2}(k) \\
&\leq \sum_{i=0}^{d} \binom{k}{i} + \sum_{i=0}^{d} \binom{k}{i} \\
&= \sum_{i=0}^{d} \binom{k}{i} + \sum_{i=0}^{d} \binom{k}{k-i} \\
&= \sum_{i=0}^{d} \binom{k}{i} + \sum_{i=k-d}^{k} \binom{k}{i} \\
&\leq \sum_{i=0}^{d} \binom{k}{i} + \sum_{i=d+2}^{k} \binom{k}{i} \\
&< \sum_{i=0}^{d} \binom{k}{i} + \sum_{i=d+1}^{k} \binom{k}{i} \\
&= \sum_{i=0}^{k} \binom{k}{i} \\
&= 2^k .
\end{aligned}$$

12. Throughout this question, we use the convention $\mathrm{sign}(0) = -1$.

(a) It suffices to show that a set $C \subseteq \mathcal{X}$ is shattered by $POS(\mathcal{F})$ if and only if it's shattered by $POS(\mathcal{F} + g)$.

Assume that a set $C = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ of size m is shattered by $POS(\mathcal{F})$. Let $(y_1, \ldots, y_m) \in \{-1, 1\}^m$. We first argue that there exists f, for which $f(\mathbf{x}_i) > 0$ if $y_i = 1$, and $f(\mathbf{x}_i) < 0$ if $y_i = -1$. To see this, choose a function $\phi \in \mathcal{F}$ such that for every $i \in [m]$, $\phi(\mathbf{x}_i) > 0$ if and only if $y_i = 1$. Similarly, choose a function $\psi \in \mathcal{F}$, such that for every $i \in [m]$, $\psi(\mathbf{x}_i) > 0$ if and only if $y_i = -1$. Than $f = \phi - \psi$ (which belongs to $\mathcal{F}$ since $\mathcal{F}$ is linearly closed) satisfies the requirements. Next, since $\mathcal{F}$ is closed under multiplication by (positive) scalars, we may assume w.l.o.g. that $|f(\mathbf{x}_i)| \geq |g(\mathbf{x}_i)|$ for every $i \in [m]$. Consequently, for every $i \in [m]$, $\text{sign}((f+g)(\mathbf{x}_i)) = \text{sign}(f(\mathbf{x}_i)) = y_i$. We conclude that C is shattered by $POS(\mathcal{F}+g)$.

Assume now that a set $C = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ of size m is shattered by $POS(\mathcal{F}+g)$. Choose $\phi \in \mathcal{F}$ such that for every $i \in [m]$, $\text{sign}((\phi+g)(\mathbf{x}_i)) = y_i$. Analogously, find $\psi \in \mathcal{F}$ such that for every $i \in [m]$, $\text{sign}((\psi+g)(\mathbf{x}_i)) = -y_i$. Let $f = \phi - \psi$. Note that $f \in \mathcal{F}$. Using the identity $\phi - \psi = (\phi+g) - (\psi+g)$, we conclude that for every $i \in [m]$, $\text{sign}(f(\mathbf{x}_i)) = y_i$. Hence, C is shattered by $POS(\mathcal{F})$.

(b) The result will follow from the following claim:

Claim 6.1. *For any positive integer d, $\tau_{POS(\mathcal{F})}(d) = 2^d$ (i.e., there exists a shattered set of size d) if and only if there exists an independent subset $\mathcal{G} \subseteq \mathcal{F}$ of size d.*

The lemma implies that the VC-dimension of $POS(\mathcal{F})$ equals to the dimension of $\mathcal{F}$. Note that both might equal ∞. Let us prove the claim.

Proof. Let $\{f_1, \ldots, f_d\} \in \mathcal{F}$ be a linearly independent set of size d. Following the hint, for each $\mathbf{x} \in \mathbb{R}^n$, let $\phi(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_d(\mathbf{x}))$. We use induction to prove that there exist $\mathbf{x}_1, \ldots, \mathbf{x}_d$ such that the set $\{\phi(\mathbf{x}_j) : j \in [d]\}$ is an independent subset of $\mathbb{R}^d$, and thus spans $\mathbb{R}^d$. The case $d = 1$ is trivial. Assume now that the argument holds for any integer $k \in [d-1]$. Apply the induction hypothesis to choose $\mathbf{x}_1, \ldots, \mathbf{x}_{d-1}$ such that the set $\{(f_1(\mathbf{x}_j), \ldots, f_{d-1}(\mathbf{x}_j)) : j \in [d-1]\}$ is independent. If there doesn't exist $\mathbf{x}_d \notin \{\mathbf{x}_j : j \in [d-1]\}$ such that $\{\phi(\mathbf{x}_j) : j \in [d]\}$ is independent, then f_d depends on $f_1, \ldots, f_{d-1}$. This would contradict our earlier assumption.

Let $\{\phi(\mathbf{x}_j) : j \in [d]\}$ be such an independent set. For each $j \in [d]$, let $\mathbf{v}_j = \phi(\mathbf{x}_j)$. Then, since $\mathcal{F}$ is linearly closed,

$$[POS(\mathcal{F})]_{\{\mathbf{x}_1,\ldots,\mathbf{x}_d\}} \supseteq \{(\text{sign}(\langle \mathbf{w}, \mathbf{v}_1 \rangle), \ldots, \text{sign}(\langle \mathbf{w}, v_d \rangle)) : \mathbf{w} \in \mathbb{R}^d\}$$

Since we can replace $\mathbf{w}$ by $V^\top \mathbf{w}$, where V is the $d \times d$ matrix whose j-th column is $\mathbf{v}_j$, we may assume w.l.o.g. that $\{\mathbf{v}_1, \ldots, \mathbf{v}_d\}$ is the standard basis, i.e. $\mathbf{v}_j = \mathbf{e}_j$ for every $j \in [d]$ (simply note that $\langle V^\top \mathbf{w}, \mathbf{e}_j\rangle = \langle \mathbf{w}, \mathbf{v}_j\rangle$). Since the standard basis is shattered by the hypothesis set of homogenous halfspaces (Theorem 9.2), we conclude that $|[POS(\mathcal{F})]_{\{\mathbf{x}_1,\ldots,\mathbf{x}_d\}}| = 2^d$, i.e. $\tau_{POS(\mathcal{F})}(d) = 2^d$.

The other direction is analogous. Assume that there doesn't exist an independent subset $\mathcal{G} \subseteq \mathcal{F}$ of size d. Hence, $\mathcal{F}$ has a basis $\{f_j : j \in [k]\}$, for some $k < d$. For each $\mathbf{x} \in \mathbb{R}^n$, let $\phi(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_k(\mathbf{x}))$. Let $\{\mathbf{x}_1, \ldots, \mathbf{x}_d\} \subseteq \mathbb{R}^n$ be a set of size d. We note that

$$[POS(\mathcal{F})]_{\{\mathbf{x}_1,\ldots,\mathbf{x}_d\}} \subseteq \{(\mathrm{sign}(\langle \mathbf{w}, \phi(\mathbf{x}_1)\rangle), \ldots, \mathrm{sign}(\langle \mathbf{w}, \phi(\mathbf{x}_d)\rangle)) : \mathbf{w} \in \mathbb{R}^d\}$$

The set $\{\phi(\mathbf{x}_j) : j \in [d]\}$ is a linearly dependent subset of $\mathbb{R}^k$. Hence, by (the opposite direction of) Theorem 9.2, the class of halfspaces doesn't shatter it. It follows that $|[POS(\mathcal{F})]_{\{\mathbf{x}_1,\ldots,\mathbf{x}_d\}}| < 2^d$, and hence $\tau_{POS(\mathcal{F})}(d) < 2^d$. □

(c) i. Let $\mathcal{F} = \{f_{\mathbf{w},b} : \mathbf{w} \in \mathbb{R}^n, b \in \mathbb{R}\}$, where $f_{\mathbf{w},b}(\mathbf{x}) = \langle \mathbf{w}, \mathbf{x}\rangle + b$. We note that $HS_n = POS(\mathcal{F})$.

ii. Let $\mathcal{F} = \{f_{\mathbf{w}} : \mathbf{w} \in \mathbb{R}^n\}$, where $f_{\mathbf{w}}(\mathbf{x}) = \langle \mathbf{w}, \mathbf{x}\rangle$. We note that $HHS_n = POS(\mathcal{F})$.

iii. Let $\mathcal{B}_n = \{h_{\mathbf{a},r} : \mathbf{a} \in \mathbb{R}^n, r \in \mathbb{R}_+\}$ be the class of open balls in $\mathbb{R}^n$. That is, $h_{\mathbf{a},r}(\mathbf{x}) = 1$ if and only if $\|\mathbf{x} - \mathbf{a}\| < r$.

We first find a dudley class representation for the class. Observe that for every $\mathbf{a} \in \mathbb{R}^n$, $r \in \mathbb{R}_+$, $h_{\mathbf{a},r}(\mathbf{x}) = POS(s_{\mathbf{a},r}(\mathbf{x}))$, where $s_{\mathbf{a},r}(\mathbf{x}) = 2\langle \mathbf{a}, \mathbf{x}\rangle - \|\mathbf{a}\|^2 + r^2 - \|\mathbf{x}\|^2$. Let $\mathcal{F}$ be the vector space of functions of the form $f_{\mathbf{a},q}(\mathbf{x}) = 2\langle \mathbf{a}, \mathbf{x}\rangle + q$, where $\mathbf{a} \in \mathbb{R}^n$, and $q \in \mathbb{R}$. Let $g(\mathbf{x}) = -\|\mathbf{x}\|^2$. The space $\mathcal{F}$ is spanned by the set $\{f_{\mathbf{e}_j} : j \in [n+1]\}$ (here we use the standard basis of $\mathbb{R}^{n+1}$). Hence, $\mathcal{F}$ is a vector space with dimensionality $n+1$. Furthermore, $POS(\mathcal{F} + g) \supseteq \mathcal{B}_n$. Following the results from the previous sections, we conclude that $\mathrm{VCdim}(\mathcal{B}_n) \le n+1$.

Next, we prove that the set $F = \{\mathbf{0}, \mathbf{e}_1, \ldots, \mathbf{e}_n\}$ (where $\mathbf{0}$ denotes the zero vector, and $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is the standard basis) is shattered. Let $E \subseteq F$. If E is empty, then pick $r = 0$ to obtain the corresponding labeling. Assume that $E \neq \emptyset$. Let $\mathbf{a} = \sum_{j:\mathbf{e}_j \in E} \mathbf{e}_j$. Note that $\|\mathbf{a} - \mathbf{0}\| = |E|$, and

$$\|\mathbf{e}_j - \mathbf{a}\|_2 = \begin{cases} |E| - 1 & \mathbf{e}_j \in E \\ |E| + 1 & \mathbf{e}_j \notin E \end{cases}$$

Now, if $\mathbf{0} \in E$, set $r = |E| + 1/2$, and otherwise, set $r = |E| - 1/2$. It follows that $h_{\mathbf{a},r}(\mathbf{x}) = \mathbb{1}_{[x \in E]}$. Hence, F is indeed shattered. We conclude that $\text{VCdim}(\mathcal{B}_n) \geq n+1$.

iv. A. Let $\mathcal{F} = \{f_{\mathbf{w}} : \mathbf{w} \in \mathbb{R}^{d+1}\}$, where $f_{\mathbf{w}}(x) = \sum_{j=1}^{d+1} w_j x^{j-1}$. Then $P_1^d = POS(\mathcal{F})$. The vector space $\mathcal{F}$ is spanned by the (functions corresponding to the) standard basis of $\mathbb{R}^{d+1}$. Hence $\dim(\mathcal{F}) = d+1$, and $\text{VCdim}(P_1^d) = d+1$.

B. We need to introduce some notation. For each $n \in \mathbb{N}$, let $\mathcal{X}_n = \mathbb{R}$. Denote the product $\prod_{n=1}^{\infty} \mathcal{X}_n$ by $\mathbb{R}^{\infty}$. Denote by $\mathcal{X}$ the subset of $\mathbb{R}^{\infty}$ containing all sequences with finitely many non-zero elements. Let $\mathcal{F} = \{f_{\mathbf{w}} : \mathbf{w} \in \mathcal{X}\}$, where $f_{\mathbf{w}}(\mathbf{x}) = \sum_{n=1}^{\infty} w_n x^{n-1}$. A basis for this space is the set of functions corresponding to the infinite standard basis of $\mathcal{X}$ (i.e., the set $\{(1,0,\ldots),(0,1,0,\ldots),\ldots\}$). Note that the set $\bigcup_{d \in \mathbb{N}} P_1^d$ of all polynomial classifiers obeys $\bigcup_{d \in \mathbb{N}} P_1^d = POS(\mathcal{F})$. Hence, we have $\text{VCdim}(\bigcup_{d \in \mathbb{N}} P_1^d) = \infty$.

C. Recall that the number of monomials of degree k in n variables equals to the number of ways to choose k elements from a set of size n, while repetition is allowed, but order doesn't matter. This number equals $\binom{n+k-1}{k}$, and it's denoted by $\left(\!\binom{n}{k}\!\right)$. For $\mathbf{x} \in \mathbb{R}^n$, we denote by $\psi(\mathbf{x})$ the vector with all monomials of degree at most d associated with $\mathbf{x}$.
Given $n, d \in \mathbb{N}$, let $\mathcal{F} = \{f_{\mathbf{w}} : \mathbf{w} \in \mathbb{R}^N\}$, where $N = \sum_{k=0}^{d} \left(\!\binom{n}{k}\!\right)$, and $f_{\mathbf{w}}(\mathbf{x}) = \sum_{i=1}^{N} w_i (\psi(\mathbf{x}))_i$. The vector space $\mathcal{F}$ is spanned by the (functions corresponding to the) standard basis of $\mathbb{R}^N$. We note that $P_n^d = POS(\mathcal{F})$. Therefore, $\text{VCdim}(P_n^d) = N$.

7 Non-uniform Learnability

1. (a) Let $n = \max_{h \in \mathcal{H}}\{|d(h)|\}$. Since each $h \in \mathcal{H}$ has a unique description, we have

$$|\mathcal{H}| \leq \sum_{i=0}^{n} 2^i = 2^{n+1} - 1 \ .$$

It follows that $\text{VCdim}(\mathcal{H}) \leq \lfloor \log |\mathcal{H}| \rfloor \leq n+1 \leq 2n$.

Fourth and fifth parts: Let $\epsilon > 0$. There exists $N \in \mathbb{N}$ such that $\sum_{n \geq N} \mathcal{D}(\{x_n\}) < \epsilon$. It follows also that $\mathcal{D}(\{x_n\}) < \epsilon$ for every $n \geq N$. Let $\eta = \mathcal{D}(\{x_N\})$. Then, $\mathcal{D}(\{x_k\}) \geq \eta$ for every $k \in [N]$. Then, by applying the union bound,

$$
\begin{aligned}
\mathop{\mathbb{P}}_{S \sim \mathcal{D}^m}[\mathcal{D}(\{x : x \notin S\}) > \epsilon] &\leq \mathop{\mathbb{P}}_{S \sim \mathcal{D}^m}[\exists i \in [N] : x_i \notin S] \\
&\leq \sum_{i=1}^{N} \mathop{\mathbb{P}}_{S \sim \mathcal{D}^m}[x_i \notin S] \\
&\leq N(1-\eta)^m \\
&\leq N \exp(-\eta m) \ .
\end{aligned}
$$

If $m \geq m_{\mathcal{D}}(\epsilon, \delta) := \left\lceil \frac{\log(N/\delta)}{\eta} \right\rceil$, then the probability above is at most δ.

Denote the algorithm memorize by M. Given $\mathcal{D}, \epsilon, \delta$, we equip the algorithm Memorize with $m_{\mathcal{D}}(\epsilon, \delta)$ i.i.d. instances. By the previous part, with probability at least $1 - \delta$, M observes all the instances in $\mathcal{X}$ (along with their labels) except a subset with probability mass at most ϵ. Since h is consistent with the elements seen by M, it follows that with probability at least $1 - \delta$, $L_{\mathcal{D}}(h) \leq \epsilon$.

8 The Runtime of Learning

1. Let $x_1 < x_2 < \ldots < x_m$ be the points in S. Let $x_0 = x_1 - 1$. To find an ERM hypothesis, it suffices to calculate, for each $i, j \in \{0, 1, \ldots, m\}$, the loss induced by picking the interval $[x_i, x_j]$, and find the minimal value. For convenience, denote the loss for each interval by $\ell_{i,j}$. A naive implementation of this strategy runs in time $\Theta(m^3)$. However, dynamic programming yields an algorithm which runs in time $O(m^2)$. The main observation is that given $\ell_{i,j}$, $\ell_{i+1,j}$ and $\ell_{i,j+1}$ can be calculated in time $O(1)$. Consider the table/matrix whose (i,j)-th value is $\ell_{i,j}$. Our algorithm starts with calculating $\ell_{1,j}$ for every j and then, fills the missing values, "row by row". Finding the optimal value among the $\ell_{i,j}$ is done in time $O(m^2)$ as well. Thus, the overall runtime of the proposed algorithm is $O(m^2)$.

2. Let $(\mathcal{H}_n)_{n \in \mathbb{N}}$ be a sequence with the properties described in the question. It follows that given a sample S of size m, the hypotheses in $\mathcal{H}_n$ can be partitioned into at most $\binom{m}{n} \leq O(m^{O(n)})$ equivalence classes, such that any two hypotheses in the same equivalence class have the

same behaviour w.r.t. S. Hence, to find an ERM, it suffices to consider one representative from each equivalent class. Calculating the empirical risk of any representative can be done in time $O(mn)$. Hence, calculating all the empirical errors, and finding the minimizer is done in time $O\left(mnm^{O(n)}\right) = O\left(nm^{O(n)}\right)$.

3. (a) Let $\mathcal{A}$ be an algorithm which implements the ERM rule w.r.t. the class HS_n. We next show that a solution of $\mathcal{A}$ implies a solution to the Max FS problem.

   Let $A \in \mathbb{R}^{m\times n}, \mathbf{b} \in \mathbb{R}^m$ be an input to the Max FS problem. Since the system requires strict inequalities, we may assume w.l.o.g. that $b_i \neq 0$ for every $i \in [m]$ (a solution to the original system exists if and only if there exists a solution to a system in which each 0 in $\mathbf{b}$ is replaced with some $\epsilon > 0$). Furthermore, since the system $A\mathbf{x} > \mathbf{b}$ is invariant under multiplication by a positive scalar, we may assume that $|b_i| = |b_j|$ for every $i, j \in [m]$. Denote $|b_i|$ by b.

   We construct a training sequence $S = ((\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_m, y_m))$ such that $\mathbf{x}_i = -\text{sign}(b_i)A_{i\rightarrow}$ (where $A_{i\rightarrow}$ denotes the i-th row of A), and $y_i = -\text{sign}(b_i)$ for every $i \in [m]$.

   We now argue that $(\mathbf{w}, b) \in \mathbb{R}^n \times \mathbb{R}$ induce an ERM solution w.r.t. S and HS_n if and only if $\mathbf{w}$ is a solution to the Max FS problem. Since minimizing the empirical risk is equivalent to maximizing the number of correct classifications, it suffices to show that $h_{\mathbf{w},b}$ classifies $\mathbf{x}_i$ correctly if and only if the i-th inequality is satisfied using $\mathbf{w}$. Indeed[4],

$$\begin{aligned} h_w(\mathbf{x}_i) = y_i &\Leftrightarrow y_i(\langle w, \mathbf{x}_i\rangle + b) > 0 \\ &\Leftrightarrow \langle w, A_{i\rightarrow}\rangle - b_i > 0\ . \end{aligned}$$

   It follows that $\mathcal{A}$ can be applied to find a solution to the Max FS problem. Hence, since Max FS is NP-hard, ERM_{HS_n} is NP-hard as well. In other words, unless $P = NP$, $\text{ERM}_{HS_n} \notin P$.

   (b) Throughout this question, we denote by $c_i \in [k]$ the color of v_i.

[4] Note that S is linearly separable if and only if it's separable with strict inequalities. That is, S is separable iff there exist $\mathbf{w}, b$ s.t. $y_i(\langle \mathbf{w}, x_i\rangle + b) > 0$ for every $i \in [m]$. While one direction is trivial, the other follows by adjusting b properly. Note however, that this claim doesn't hold for the class of homogenous halfspaces.

Following the hints, we show that $S(G)$ is separable if and only if G is k-colorable.

First, assume that $S(G)$ is (strictly) separable, using $((\mathbf{w}_1, b_1), \ldots, (\mathbf{w}_k, b_k)) \in (\mathbb{R}^n \times \mathbb{R})^k$. That is, for every $i \in [m]$, $y_i = 1$ if and only if for every $j \in [k]$, $\langle \mathbf{w}_j, \mathbf{x}_i \rangle + b_j > 0$. Consider some vertex $v_i \in V$, and its associated instance $\mathbf{e}_i$. Since the corresponding label is negative, the set $C_i := \{j \in [k] : w_{j,i} + b_j < 0\}$ [5] is not empty. Next, consider some edge $(v_s, v_t) \in E$, and its associated instance $(\mathbf{e}_s + \mathbf{e}_t)/2$. Since the corresponding label is positive, $C_s \cap C_t = \emptyset$. It follows that we can safely pick $c_i \in C_i$ for every $i \in [n]$. In other words, G is k-colorable.

For the other direction, assume that G is k-colorable using $(c_1, \ldots, c_n)$. We associate a vector $\mathbf{w}_i \in \mathbb{R}^n$ with each color $i \in [k]$. Define $\mathbf{w}_i = \sum_{j \in [n]} -\mathbb{1}_{[c_j = i]} \mathbf{e}_j$. Let $b_1 = \ldots = b_k = b := 0.6$. Consider some vertex $v_i \in V$, and its associated instance $\mathbf{e}_i$. We note that $\mathrm{sign}(\langle \mathbf{w}_{c_i}, \mathbf{e}_i \rangle + b) = \mathrm{sign}(-0.4) = -1$, so $\mathbf{e}_i$ is classified correctly. Consider any edge (v_s, v_t). We know that $c_s \neq c_t$. Hence, for every $j \in [k]$, either $w_{j,s} = 0$ or $w_{j,t} = 0$. Hence, $\forall j \in [k]$, $\mathrm{sign}(\langle \mathbf{w}_j, (\mathbf{e}_s + \mathbf{e}_t)/2 \rangle + b_j) = \mathrm{sign}(0.1) = 1$, so the (instance associated with the) edge (v_s, v_t) is classified correctly. All in all, $S(G)$ is separable using $((\mathbf{w}_1, b_1), \ldots, (\mathbf{w}_k, b_k))$.

Let $\mathcal{A}$ be an algorithm which implements the ERM rule w.r.t. $\mathcal{H}_k^n$. Then $\mathcal{A}$ can be used to solve the k-coloring problem. Given a graph G, $\mathcal{A}$ returns "Yes" if and only if the resulting training sequence $S(G)$ is separable. Since the k-coloring problem is NP-hard (for $k \geq 3$), we conclude that $\mathrm{ERM}_{\mathcal{H}_k^n}$ is NP-hard as well.

4. We will follow the hint[6]. With probability at least $1 - 0.3 = 0.7 \geq 0.5$, the generalization error of the returned hypothesis is at most $1/|S|$. Since the distribution is uniform, this implies that the returned hypothesis is consistent with the training set.

[5] recall that $w_{j,i}$ denotes the i-th element of the j-th vector.

[6] Errata:

- The realizable assumption should be made for the task of minimizing the ERM.
- The definition of RP should be modified. We don't consider a decision problem ("Yes"/"No" question) but an optimization problem (predicting correctly the labels of the instances in the training set).

9 Linear Predictors

1. Define a vector of auxiliary variables $s = (s_1, \ldots, s_m)$. Following the hint, minimizing the empirical risk is equivalent to minimizing the linear objective $\sum_{i=1}^{m} s_i$ under the following constraints:

$$(\forall i \in [m]) \quad \mathbf{w}^T \mathbf{x}_i - s_i \leq y_i \quad , \quad -\mathbf{w}^T \mathbf{x}_i - s_i \leq -y_i \tag{3}$$

It is left to translate the above into matrix form. Let $A \in \mathbb{R}^{2m \times (m+d)}$ be the matrix $A = [X \ \ -I_m; -X \ \ -I_m]$, where $X_{i\rightarrow} = x_i$ for every $i \in [m]$. Let $\mathbf{v} \in \mathbb{R}^{d+m}$ be the vector of variables $(w_1, \ldots, w_d, s_1, \ldots, s_m)$. Define $\mathbf{b} \in \mathbb{R}^{2m}$ to be the vector $\mathbf{b} = (y_1, \ldots, y_m, -y_1, \ldots, -y_m)^T$. Finally, let $\mathbf{c} \in \mathbb{R}^{d+m}$ be the vector $\mathbf{c} = (\mathbf{0}_d; \mathbf{1}_m)$. It follows that the optimization problem of minimizing the empirical risk can be expressed as the following LP:

$$\begin{aligned} \min \quad & c^T \mathbf{v} \\ \text{s.t.} \quad & A\mathbf{v} \leq \mathbf{b} \ . \end{aligned}$$

2. Consider the matrix $X \in \mathbb{R}^{d \times m}$ whose columns are $x_1, \ldots, x_m$. The rank of X is equal to the dimension of the subspace $\mathrm{span}(\{x_1, \ldots, x_m\})$. The SVD theorem (more precisely, Lemma C.4) implies that the rank of X is equal to the rank of $A = XX^\top$. Hence, the set $\{x_1, \ldots, x_m\}$ spans $\mathbb{R}^d$ if and only if the rank of $A = XX^\top$ is d, i.e., iff $A = XX^\top$ is invertible.

3. Following the hint, let $d = m$, and for every $i \in [m]$, let $\mathbf{x}_i = \mathbf{e}_i$. Let us agree that $\mathrm{sign}(0) = -1$. For $i = 1, \ldots, d$, let $y_i = 1$ be the label of x_i. Denote by $\mathbf{w}^{(t)}$ the weight vector which is maintained by the Perceptron. A simple inductive argument shows that for every $i \in [d]$, $\mathbf{w}_i = \sum_{j<i} \mathbf{e}_j$. It follows that for every $i \in [d]$, $\langle \mathbf{w}^{(i)}, \mathbf{x}_i \rangle = 0$. Hence, all the instances $\mathbf{x}_1, \ldots, \mathbf{x}_d$ are misclassified (and then we obtain the vector $w = (1, \ldots, 1)$ which is consistent with $x_1, \ldots, x_m$). We also note that the vector $\mathbf{w}^\star = (1, \ldots, 1)$ satisfies the requirements listed in the question.

4. Consider all positive examples of the form $(\alpha, \beta, 1)$, where $\alpha^2 + \beta^2 + 1 \leq R^2$. Observe that $\mathbf{w}^\star = (0, 0, 1)$ satisfies $y\langle \mathbf{w}^\star, \mathbf{x} \rangle \geq 1$ for all such $(\mathbf{x}, y)$. We show a sequence of R^2 examples on which the Perceptron makes R^2 mistakes.

The idea of the construction is to start with the examples $(\alpha_1, 0, 1)$ where $\alpha_1 = \sqrt{R^2 - 1}$. Now, on round t let the new example be such that the following conditions hold:

(a) $\alpha^2 + \beta^2 + 1 = R^2$

(b) $\langle \mathbf{w}_t, (\alpha, \beta, 1) \rangle = 0$

As long as we can satisfy both conditions, the Perceptron will continue to err. We'll show that as long as $t \leq R^2$ we can satisfy these conditions.

Observe that, by induction, $\mathbf{w}^{(t-1)} = (a, b, t-1)$ for some scalars a, b. Observe also that $\|\mathbf{w}_{t-1}\|^2 = (t-1)R^2$ (this follows from the proof of the Perceptron's mistake bound, where inequalities hold with equality). That is, $a^2 + b^2 + (t-1)^2 = (t-1)R^2$.

W.l.o.g., let us rotate $\mathbf{w}^{(t-1)}$ w.r.t. the z axis so that it is of the form $(a, 0, t-1)$ and we have $a = \sqrt{(t-1)R^2 - (t-1)^2}$. Choose

$$\alpha = -\frac{t-1}{a} \; .$$

Then, for every β,

$$\langle (a, 0, t-1), (\alpha, \beta, 1) \rangle = 0 \; .$$

We just need to verify that $\alpha^2 + 1 \leq R^2$, because if this is true then we can choose $\beta = \sqrt{R^2 - \alpha^2 - 1}$. Indeed,

$$\begin{aligned} \alpha^2 + 1 &= \frac{(t-1)^2}{a^2} + 1 = \frac{(t-1)^2}{(t-1)R^2 - (t-1)^2} + 1 = \frac{(t-1)R^2}{(t-1)R^2 - (t-1)^2} \\ &= R^2 \frac{1}{R^2 - (t-1)} \\ &\leq R^2 \end{aligned}$$

where the last inequality assumes $R^2 \geq t$.

5. Since for every $w \in \mathbb{R}^d$, and every $x \in \mathbb{R}^d$, we have

$$\mathrm{sign}(\langle w, x \rangle) = \mathrm{sign}(\langle \eta w, x \rangle) \; ,$$

we obtain that the modified Perceptron and the Perceptron produce the same predictions. Consequently, both algorithms perform the same number of iterations.

6. In this question we will denote the class of halfspaces in $\mathbb{R}^{d+1}$ by $\mathcal{L}_{d+1}$.

   (a) Assume that $A = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\} \subseteq \mathbb{R}^d$ is shattered by $\mathcal{B}_d$. Then, $\forall \mathbf{y} = (y_1, \ldots, y_d) \in \{-1, 1\}^d$ there exists $B_{\mu,r} \in \mathcal{B}$ s.t. for every i

   $$B_{\mu,r}(\mathbf{x}_i) = y_i \ .$$

   Hence, for the above μ and r, the following identity holds for every $i \in [m]$:

   $$\operatorname{sign}\left((2\mu; -1)^T(\mathbf{x}_i; \|\mathbf{x}_i\|^2) - \|\mu\|^2 + r^2\right) = y_i \ , \tag{4}$$

   where ; denotes vector concatenation. For each $i \in [m]$, let $\phi(x_i) = (x_i; \|x\|_i^2)$. Define the halfspace $h \in \mathcal{L}_{d+1}$ which corresponds to $w = (2\mu; -1)$, and $b = \|\mu\|^2 - r^2$. Equation (4) implies that for every $i \in [m]$,

   $$h(x_i) = y_i$$

   All in all, if $A = \{x_1, \ldots, x_m\}$ is shattered by $\mathcal{B}$, then $\phi(A) := \{\phi(x_1), \ldots, \phi(x_m)\}$, is shattered by $\mathcal{L}$. We conclude that $d + 2 = \mathrm{VCdim}(\mathcal{L}_{d+1}) \geq VC(\mathcal{B}_d)$.

   (b) Consider the set C consisting of the unit vectors $\mathbf{e}_1, \ldots, \mathbf{e}_d$, and the origin $\mathbf{0}$. Let $A \subseteq C$. We show that there exists a ball such that all the vectors in A are labeled positively, while the vectors in $C \setminus A$ are labeled negatively. We define the center $\mu = \sum_{e \in A} e$. Note that for every unit vector in A, its distance to the center is $\sqrt{|A| - 1}$. Also, for every unit vector outside A, its distance to the center is $\sqrt{|A| + 1}$. Finally, the distance of the origin to the center is $\sqrt{A}$. Hence, if $0 \in A$, we will set $r = \sqrt{|A| - 1}$, and if $0 \notin A$, we will set $r = \sqrt{|A|}$. We conclude that the set C is shattered by $\mathcal{B}_d$. All in all, we just showed that $\mathrm{VCdim}(\mathcal{B}_d) \geq d + 1$.

10 Boosting

1. Let $\epsilon, \delta \in (0, 1)$. Pick k "chunks" of size $m_{\mathcal{H}}(\epsilon/2)$. Apply A on each of these chunks, to obtain $\hat{h}_1, \ldots, \hat{h}_k$. Note that the probability that $\min_{i \in [k]} L_{\mathcal{D}}(\hat{h}_i) \leq \min L_{\mathcal{D}}(h) + \epsilon/2$ is at least $1 - \delta_0^k \geq 1 - \delta/2$. Now, apply an ERM over the class $\hat{\mathcal{H}} := \{\hat{h}_1, \ldots, \hat{h}_k\}$ with the training data being the last chunk of size $\left\lceil \frac{2\log(4k/\delta)}{\epsilon^2} \right\rceil$. Denote the output hypothesis

by $\hat{h}$. Using Corollary 4.6, we obtain that with probability at least $1-\delta/2$, $L_{\mathcal{D}}(\hat{h}) \le \min_{i\in[k]} L_{\mathcal{D}}(h_i) + \epsilon/2$. Applying the union bound, we obtain that with probability at least $1-\delta$,

$$\begin{aligned} L_{\mathcal{D}}(\hat{h}) &\le \min_{i\in[k]} L_{\mathcal{D}}(h_i) + \epsilon/2 \\ &\le \min_{h\in\mathcal{H}} L_{\mathcal{D}}(h) + \epsilon \ . \end{aligned}$$

2. Note[7] that if $g(x)=1$, then $h(x)$ is either 0.5 or 1.5, and if $g(x)=-1$, then $h(x)$ is either -0.5 or -1.5.

3. The definition of ϵ_t implies that

$$\begin{aligned} \sum_{i=1}^{m} D_i^{(t)} \exp(-w_t y_i h_t(x_i)) \cdot \mathbb{1}_{[h_t(x_i)\neq y_i]} &= \epsilon_t \cdot \sqrt{\frac{1-\epsilon_t}{\epsilon_t}} \\ &= \sqrt{\epsilon_t(1-\epsilon_t)} \ . \end{aligned} \tag{5}$$

Similarly,

$$\begin{aligned} \sum_{j=1}^{m} D_j^{(t)} \cdot \exp(-w_t y_j h_t(x_j)) &= \epsilon_t \cdot \sqrt{\frac{1-\epsilon_t}{\epsilon_t}} + (1-\epsilon_t)\cdot\sqrt{\frac{\epsilon_t}{1-\epsilon_t}} \\ &= 2\sqrt{\epsilon_t(1-\epsilon_t)} \ . \end{aligned} \tag{6}$$

The desired equality is obtained by observing that the error of h_t w.r.t. the distribution D_{t+1} can be obtained by dividing Equation (5) by Equation (6).

4. (a) Let $\mathcal{X}$ be a finite set of size n. Let B be the class of all functions from $\mathcal{X}$ to $\{0,1\}$. Then, $L(B,T)=B$, and both are finite. Hence, for any $T\ge 1$,

$$\mathrm{VCdim}(B) = \mathrm{VCdim}(L(B,T)) = \log 2^n = n \ .$$

(b)
- Denote by $\mathcal{B}$ the class of decision stumps in $\mathbb{R}^d$. Formally, $\mathcal{B} = \{h_{j,b,\theta} : j\in[d], b\in\{-1,1\}, \theta\in\mathbb{R}\}$, where $h_{j,b,\theta}(\mathbf{x}) = b\cdot\mathrm{sign}(\theta - x_j)$.
  For each $j\in[d]$, let $\mathcal{B}_j = \{h_{b,\theta} : b\in\{-1,1\}, \theta\in\mathbb{R}\}$, where $h_{b,\theta}(\mathbf{x}) = b\cdot\mathrm{sign}(\theta-x_j)$. Note that $\mathrm{VCdim}(\mathcal{B}_j)=2$.
  Clearly, $\mathcal{B} = \bigcup_{j=1}^{d}\mathcal{B}_j$. Applying Exercise 11, we conclude that

$$\mathrm{VCdim}(\mathcal{B}) \le 16 + 2\log d \ .$$

[7]Errata: w_1 should be -0.5

- Assume w.l.o.g. that $d = 2^k$ for some $k \in \mathbb{N}$ (otherwise, replace d by $2^{\lfloor \log d \rfloor}$). Let $A \in \mathbb{R}^{k \times d}$ be the matrix whose columns range over the (entire) set $\{0,1\}^k$. For each $i \in [k]$, let $\mathbf{x}_i = A_{i\rightarrow}$. We claim that the set $C = \{\mathbf{x}_i, \ldots, \mathbf{x}_k\}$ is shattered. Let $I \subseteq [k]$. We show that we can label the instances in I positively, while the instances $[k] \backslash I$ are labeled negatively. By our construction, there exists an index j such that $A_{i,j} = \mathbf{x}_{i,j} = 1$ iff $i \in I$. Then, $h_{j,-1,1/2}(\mathbf{x}_i) = 1$ iff $i \in I$.

(c) Following the hint, for each $i \in [Tk/2]$, let $\mathbf{x}_i = \lceil i/k \rceil A_{i,\rightarrow}$. We claim that the set $C = \{\mathbf{x}_i : i \in [Tk/2]\}$ is shattered by $L(B_d, T)$. Let $I \subseteq [Tk/2]$. Then $I = I_1 \cup \ldots \cup I_{T/2}$, where each I_t is a subset of $\{(t-1)k+1, \ldots, tk\}$. For each $t \in [T/2]$, let j_t be the corresponding column of A (i.e., $A_{i,j} = 1$ iff $(t-1)k + i \in I_t$). Let

$$\begin{aligned} h(x) = \operatorname{sign}((&h_{j_1,-1,1/2} + h_{j_1,1,3/2} + h_{j_2,-1,3/2} + h_{j_2,1,5/2} + \ldots \\ &+ h_{j_{T/2-1},-1,T/2-3/2} + h_{j_{T/2-1},1,T/2-1/2} + h_{j_{T/2},-`1,T/2-1/2})(x)) \ . \end{aligned}$$

Then $h(\mathbf{x}_i) = 1$ iff $i \in I$. Finally, observe that $h \in L(B_d, T)$.

5.
- We apply dynamic programming. Let $A \in \mathbb{R}^{n \times n}$. First, observe that filling sequentially each item of the first column and the first row of $I(A)$ can be done in a constant time. Next, given $i, j \geq 2$, we note that $I(A)_{i,j} = I(A)_{i-1,j} + I(A)_{i,j-1} - I(A)_{i-1,j-1}$. Hence, filling the values of $I(A)$, row by row, can be done in time linear in the size of $I(A)$.
- We show the calculation of type B. The calculation of the other types is done similarly. Given $i_1 < i_2, j_1 < j_2$, consider the following rectangular regions:

$$\begin{aligned} A_U &= \{(i,j) : i_1 \leq i \leq \lfloor (i_2 + i_1)/2 \rfloor, j_1 \leq j \leq j_2\} \ , \\ A_D &= \{(i,j) : \lceil (i_2 + i_1)/2 \rceil \leq i \leq i_2, j_1 \leq j \leq j_2\} \ . \end{aligned}$$

Let $b \in \{-1, 1\}$. Let h be the associated hypothesis. That is, $h(A) = b\left(\sum_{(i,j) \in A_L} A_{i,j} - \sum_{(i,j) \in A_R} A_{i,j}\right)$. Let $i_3 = \lfloor (i_1 + i_2)/2 \rfloor, i_4 = \lceil (i_1 + i_2)/2 \rceil$. Then,

$$\begin{aligned} h(A) = &B_{i_2,j_2} - B_{i_3,j_2} - B_{i_2,j_1-1} + B_{i_3,j_1-1} \\ &- (B_{i_3,j_2} - B_{i_1-1,j_2} - B_{i_3,j_1-1} + B_{i_1-1,j_1-1}) \end{aligned}$$

We conclude that $h(A)$ can be calculated in constant time given $B = I(A)$.

3. **Perceptron as a sub-gradient descent algorithm:**

   - Clearly, $f(\mathbf{w}^\star) \leq 0$. If there is strict inequality, then we can decrease the norm of $\mathbf{w}^\star$ while still having $f(\mathbf{w}^\star) \leq 0$. But $\mathbf{w}^\star$ is chosen to be of minimal norm and therefore equality must hold. In addition, any $\mathbf{w}$ for which $f(\mathbf{w}) < 1$ must satisfy $1 - y_i\langle \mathbf{w}, \mathbf{x}_i\rangle < 1$ for every i, which implies that it separates the examples.
   - A sub-gradient of f is given by $-y_i\mathbf{x}_i$, where $i \in \text{argmax}\{1 - y_i \langle \mathbf{w}, \mathbf{x}_i\rangle\}$.
   - The resulting algorithm initializes $\mathbf{w}$ to be the all zeros vector and at each iteration finds $i \in \text{argmin}_i\, y_i\langle \mathbf{w}, \mathbf{x}_i\rangle$ and update $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} + \eta y_i \mathbf{x}_i$. The algorithm must have $f(\mathbf{w}^{(t)}) < 0$ after $\|\mathbf{w}^\star\|^2 R^2$ iterations. The algorithm is almost identical to the Batch Perceptron algorithm with two modifications. First, the Batch Perceptron updates with any example for which $y_i\langle \mathbf{w}^{(t)}, \mathbf{x}_i\rangle \leq 0$, while the current algorithm chooses the example for which $y_i\langle \mathbf{w}^{(t)}, \mathbf{x}_i\rangle$ is minimal. Second, the current algorithm employs the parameter η. However, it is easy to verify that the algorithm would not change if we fix $\eta = 1$ (the only modification is that $\mathbf{w}^{(t)}$ would be scaled by $1/\eta$).

4. **Variable step size:** The proof is very similar to the proof of Theorem 14.11. Details can be found, for example, in [1].

15 Support Vector Machines

1. Let $\mathcal{H}$ be the class of halfspaces in $\mathbb{R}^d$, and let $S = ((\mathbf{x}_i, y_i))_{i=1}^m$ be a linearly separable set. Let $\mathcal{G} = \{(\mathbf{w}, b) : \|w\| = 1, (\forall i \in [m])\ y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) > 0\}$. Our assumptions imply that this set is non-empty. Note that for every $(\mathbf{w}, b) \in \mathcal{G}$,

   $$\min_{i\in[m]} y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) > 0 \ .$$

   On the contrary, for every $(\mathbf{w}, b) \notin \mathcal{G}$,

   $$\min_{i\in[m]} y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) \leq 0 \ .$$

   It follows that

   $$\arg \max_{\substack{(\mathbf{w},b):\\ \|w\|=1}} \min_{i\in[m]} y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) \subseteq \mathcal{G} \ .$$

Hence, solving the second optimization problem is equivalent to the following optimization problem:

$$\arg \max_{(\mathbf{w},b)\in\mathcal{G}} \min_{i\in[m]} y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) \ .$$

Finally, since for every $(\mathbf{w}, b) \in \mathcal{G}$, and every $i \in [m]$, $y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) = |\langle \mathbf{w}, \mathbf{x}_i\rangle + b|$, we obtain that the second optimization problem is equivalent to the first optimization problem.

2. Let $S = ((x_i, y_i))_{i=1}^m \subseteq (\mathbb{R}^d \times \{-1, 1\})^m$ be a linearly separable set with a margin γ, such that $\max_{i\in[m]} \|x_i\| \ \leq \rho$ for some $\rho > 0$. The margin assumption implies that there exists $(\mathbf{w}, b) \in \mathbb{R}^d \times \mathbb{R}$ such that $\|w\| = 1$, and
$$(\forall i \in [m]) \qquad y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) \geq \gamma \ .$$
Hence,
$$(\forall i \in [m]) \qquad y_i(\langle \mathbf{w}/\gamma, \mathbf{x}_i\rangle + b/\gamma) \geq 1 \ .$$
Let $w^\star = \mathbf{w}/\gamma$. We have $\|\mathbf{w}^\star\| = 1/\gamma$. Applying Theorem 9.1, we obtain that the number of iterations of the perceptron algorithm is bounded above by $(\rho/\gamma)^2$.

3. The claim is wrong. Fix some integer $m > 1$ and $\lambda > 0$. Let $\mathbf{x}_0 = (0, \alpha) \in \mathbb{R}^2$, where $\alpha \in (0, 1)$ will be tuned later. For $k = 1, \ldots, m-1$, let $\mathbf{x}_k = (0, k)$. Let $y_0 = \ldots = y_{m-1} = 1$. Let $S = \{(\mathbf{x}_i, y_i) : i \in \{0, 1, \ldots, m-1\}\}$. The solution of hard-SVM is $\mathbf{w} = (0, 1/\alpha)$ (with value $1/\alpha^2$). However, if
$$\lambda \cdot 1 + \frac{1}{m}(1 - \alpha) \leq \frac{1}{\alpha^2} \ ,$$
the solution of soft-SVM is $\mathbf{w} = (0, 1)$. Since $\alpha \in (0, 1)$, it suffices to require that $\frac{1}{\alpha^2} > \lambda + 1/m$. Clearly, there exists $\alpha_0 > 0$ s.t. for every $\alpha < \alpha_0$, the desired inequality holds. Informally, if α is small enough, then soft-SVM prefers to "neglect" $\mathbf{x}_0$.

4. Define the function $g : \mathcal{X} \to \mathbb{R}$ by $g(x) = \max_{y\in\mathcal{Y}} f(x, y)$. Clearly, for every $x \in \mathcal{X}$ and every $y \in \mathcal{Y}$,
$$g(x) \geq f(x, y) \ .$$
Hence, for every $y \in \mathcal{Y}$,
$$\min_{x\in\mathcal{X}} \max_{y\in\mathcal{Y}} f(x, y) = \min_{x\in\mathcal{X}} g(x) \geq \min_{x\in\mathcal{X}} f(x, y) \ .$$